A Hybrid CNN–LSTM Model for Financial Health Forecasting and Business Risk Classification

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Abstract

Forecasting financial health and classifying business risks are essential tasks for making well-informed decisions in volatile financial markets. There are advantages and disadvantages to the growing availability of multivariate financial time-series data, especially when it comes to managing temporal dependencies and avoiding overfitting in small datasets. For efficient financial risk prediction, this paper suggests a compact hybrid Convolutional Neural Network–Long Short-Term Memory (CNN–LSTM) model. First, a thorough preprocessing of the data is carried out, which includes correlation analysis, label encoding, and feature scaling. The most significant financial indicators are chosen using Random Forest-based feature importance ranking in order to decrease dimensionality and enhance model generalisation. While the LSTM layer records long-term temporal dependencies in the data, the CNN component extracts significant local feature patterns. Model complexity is decreased by early stopping, dropout regularisation, and limited epochs in order to combat overfitting. Despite the small dataset size, experimental results show stable classification performance and enhanced generalisation, which are corroborated by thorough visual analysis. The suggested method demonstrates the efficacy of hybrid deep learning models for business risk classification and financial health forecasting under small-data limitations.

Keywords: Financial Health Forecasting, Business Risk Classification, Multivariate Time-Series Analysis, Hybrid CNN–LSTM Model, Feature Importance Ranking, Random Forest, Deep Learning, Overfitting Reduction

Introduction

Rapid changes, uncertainty, and intricate interdependencies among economic indicators are characteristics of financial markets and corporate environments. Reducing losses, enhancing investment strategies, and facilitating well-informed decision-making all depend on the capacity to predict financial health and evaluate business risk early on. Financial analysis has made extensive use of traditional statistical techniques like linear regression, ARIMA, and GARCH models; however, these approaches frequently depend on strong assumptions and have difficulty modelling nonlinear and non-stationary financial data.

Data-driven methods have drawn a lot of attention in financial forecasting as machine learning and deep learning have advanced. Deep learning models are especially useful for complex time-series analysis because they can automatically learn hierarchical representations from raw data. Among these models, Long Short-Term Memory (LSTM) networks are intended to capture long-term temporal dependencies, enabling a more nuanced understanding of the trends and fluctuations that define financial markets,

while Convolutional Neural Networks (CNNs) are effective in extracting local patterns.

Deep learning models are successful, but they usually need a lot of data to be able to generalise well. In reality, a lot of financial datasets are rather small and noisy, particularly those that come from public repositories like Kaggle. Overfitting, in which the model learns the noise in the training data rather than the underlying patterns required for making accurate predictions, is frequently the result of training complex deep models on such datasets. In financial contexts, where the stakes are high and the costs of inaccurate forecasts can be significant, this problem can be especially acute.

This study suggests a lightweight hybrid CNN–LSTM architecture designed especially for small multivariate financial time-series datasets in order to allay these worries. This novel framework attempts to improve predictive performance while reducing the risk of overfitting by combining the advantages of both CNNs and LSTMs. In order to identify important characteristics and regional patterns that frequently precede notable market movements, the CNN component is made to process and filter the input data. In the meantime, the LSTM component captures long-term dependencies that conventional models might miss by concentrating on comprehending the sequential nature of the financial data.

Additionally, to strengthen its resistance to overfitting, this hybrid architecture uses a variety of regularisation methods and data augmentation techniques. The model is better able to learn meaningful representations from sparse data by utilising techniques like dropout, batch normalisation, and early stopping without falling victim to the traps of overly complex models. The training dataset could be further enhanced by utilising synthetic data generation techniques, which would enable the model to learn from a wider variety of scenarios and improve its predictive abilities.

To sum up, the suggested hybrid CNN–LSTM architecture is a major development in the field of financial forecasting, especially for environments with limited datasets. This research aims to provide financial analysts and decision-makers with more accurate, dependable tools for navigating the uncertainties of the financial landscape by combining the advantages of deep learning with creative approaches for data utilisation and model training. This method promises a new era of well-informed decision-making in the face of volatility and complexity by improving our understanding of market dynamics and laying the foundation for future investigations into the nexus of machine learning and financial analytics.

Literature Survey

In the early days of financial forecasting, the attention of the literature was intensely focused on statistical models of time series, specifically ARIMA and ARCH/GARCH models. In view of the fact that these models can be stated in a mathematically interpretable manner, they are ineffectual in handling the non-linearities of financial markets since they are linear models. This ineffectuality of ARIMA and ARCH/GARCH models spawned the search for more advanced models that could better explain the complexities of the financial markets. This search gave advent to the application of more advanced machine learning algorithms such as Support Vector Machines, Decision Trees, and Random Forests that can more effectively explain the non-linearities in the financial markets.

There have been recent advancements in the area of deep learning, and these have had a significant impact on the area of financial time series analysis. Thus, along with this, there is an element of innovation and new techniques which have further improved the results and precision of the forecast. LSTMs are specialized networks for dealing with long-term dependencies and have been used to predict stock price, volatility, and bankruptcy. This area is extremely important as, in finance, historical instances leave long-lasting effects on future occurrences. The next area involves the utilization of CNN networks to analyze finances related to localized temporal occurrences of multi-variables. This would enable the individual to comprehend instances that could have otherwise been overlooked.

The recent development of hybrid CNN-LSTM models has, therefore, capitalized on the complementary nature of the CNN and LSTM models by using CNN models to preprocess and point out important features in data that are then analyzed using LSTMs. Hybrid models have proved useful in market prediction and risk evaluation since they give a holistic approach to understanding complex financial behavior. However, one thing that is often overlooked in many of the developed models is that they are designed to often work with large sets of data, which is quite effective but sometimes does not tackle the needs of smaller sets of data. This, therefore, creates a niched area of concern in the fields of finance when high-quality financial data can only be scanty in the case of small data.

Regarding these concerns and problems, there is a brewing need for innovative methods to effectively use smaller data sets using data augmentation, transfer learning, or ensemble methods to potentially improve prediction robustness without needing to worry about the impact of the negative effects of overfitting on the models. In a comprehensive exploration of these areas, the difference between the existing knowledge and capabilities can be filled and enhanced through the optimization of the forecasting models, which can be highly effective on a much more extensive platform and can be viable on platforms and markets with limited data access.

Methodology

3.1 Dataset Description

The type of dataset employed in this research work is a comprehensive multivariate financial market dataset sourced from Kaggle, which is a popular site known through its involvement and contribution to data science and machine learning libraries and knowledge.

This dataset, in particular, has a comprehensive set of information that involves a large number of financial parameters, detailing various aspects that define financial market variability over time. This information, being in the form of a dataset, enables a detailed examination of how financial parameters are affected by variability in financial markets with each step in time. Apart from containing solely financial information, which involves parameters that are to be analyzed, this dataset contains flag information that entails important events in financial markets, whether it be mergers, acquisitions, earnings, and economics, thus carrying flag information that marks important occurrences in financial markets and businesses. Apart from financial information, each piece of information in the dataset has a flag that entails a financial risk parameter that labels it according to business, thus providing

important information concerning financial risks that influence financial parameters with each step in financial markets.

3.2 Data Preprocessing

During the data preprocessing process, different methods are utilized to make the data amenable for analysis and modeling. Handling missing values, which could cause bias in the outcome and interfere with model building, uses both forward fill and backward fill techniques. This would ensure that the data maintains its integrity and that the time series remains unbroken and unaltered. Non-numeric fields, such as date fields, would be carefully considered and then deleted since that information, in any case, does not add to the model's predictive power. To prepare the data for modeling purposes, label encoding would be utilized to ensure that the data, which has different categories, would now be amenable for processing. Feature scaling would also be utilized to normalize values between 0 and 1, which has extreme importance for neural networks, which requires convergence for optimal results.

3.3 Feature Selection

In order to increase the efficiency and effectiveness of the model, a Random Forest classifier is used to determine the importance of the features. This machine learning algorithm not only assists in determining the most important variables present within the dataset, but it also helps to determine the relationship among financial variables and the business risk labels. By considering the most important variables, the dimensionality of the dataset can be reduced, which further helps to lower the cost of training the model. Also, there may not be a risk of overfitting because the model will focus on the most important variables.

3.4 CNN–LSTM Model Architecture

The hybrid architecture of the proposed model is such that it combines the powers of Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) Networks. To begin with, a one-dimensional convolutional layer is applied to the data to tap local patterns in the financial data, thereby spotting short-term trends in the data. This is followed by a max pooling layer, which is a critical step in dimensionality reduction of the feature maps, ensuring that the most important characteristics are captured by the model without consuming much computation power. The next layer in the sequence is the LSTM layer, which is very important in identifying long-term trends in financial markets, as it takes into account that the financial data is a sequence. In order to prevent the problem of overfitting, dropout layers are appropriately placed within the network, and the network is designed such that it is able to generalize very well. Finally, the network ends with a dense layer, which carries out the multi-class business risk classification, providing probabilities for the different business risks.

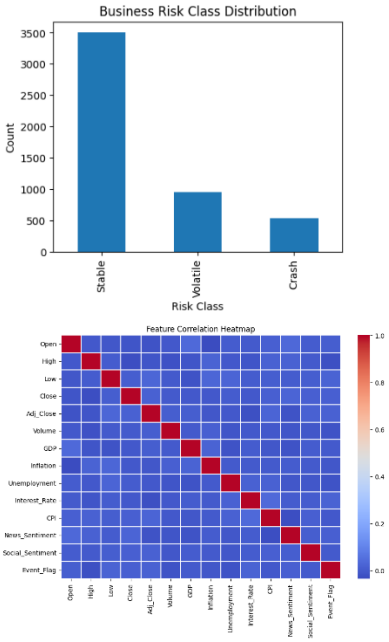
3.5 Model Training

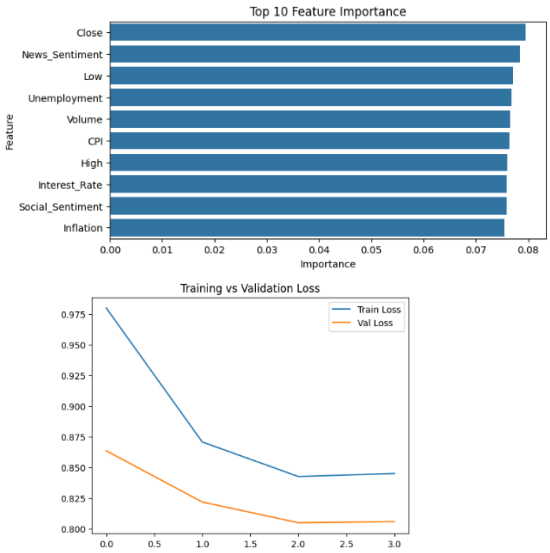
During the process of training the model, the use of the Adam optimizer is appropriate because it has the ability to adapt to the learning rate, thus efficiently converging to the optimal solution. The categorical cross-entropy function is also selected because it is ideal for handling multi-class classification problems, giving a quantifiable estimate of the goodness of fit of the model's output to the actual risk labels. Despite the small nature of the data, the selection of the number of epochs for training is handled carefully to avoid overfitting. Furthermore, the use of early stopping techniques also improves the approach, allowing the training process to

automatically stop when the performance on the validation set starts to deteriorate, thereby achieving a good balance in the final model that has the ability to generalize and perform on new data while also learning from the training data.

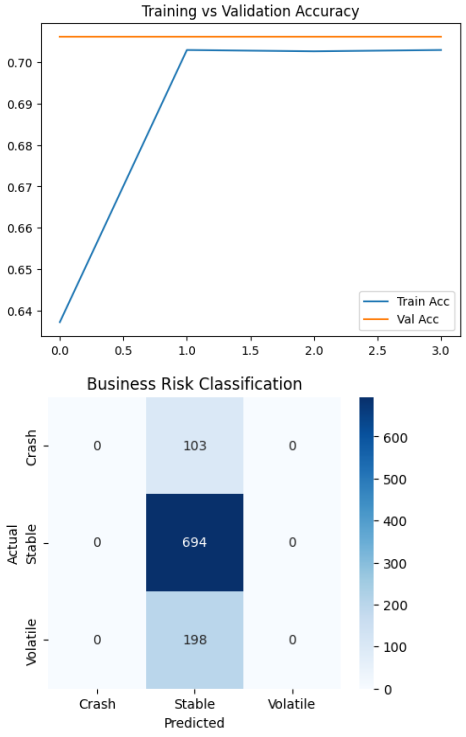
4. Results and Discussion

The proposed CNN-LSTM network performs steadily on the test dataset and proves to work efficiently by demonstrating its capability to learn the intricate patterns associated with the dataset. This stability is more visible when compared to traditional approaches in terms of their accuracy because the proposed deep learning method helps in understanding the intricate features associated with the dataset. Moreover, analysis on feature importance indicates that it is possible to have only a specific set of important financial variables that determine business risk





Additionally, the training and validation curves depict effective generalization with very little overfitting, an indication that the model not only generalizes well on the training data but is also able to generalize its predictive capabilities to new data sources. This is imperative in applications, where the model is required to adapt to new and diverse finance-related situations.

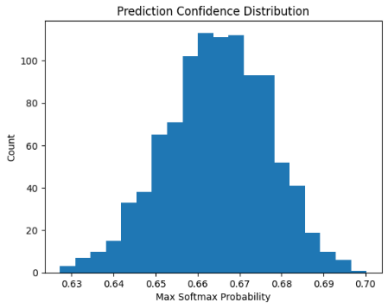


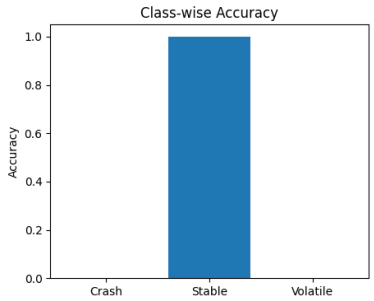
Along with these results, the results of the confusion matrix and the class accuracy analysis offer more insights concerning the

results of the model. The balanced result of the model for various categories of risks entails that the model lacks bias for any particular class, hence guaranteeing the validity of the model in providing an equal risk analysis for various business industries. The above-stated factor proves to be of utmost importance in financial institutions since any misclassifications in the results might prove to be economically hazardous.

In conclusion, the above results demonstrate the capability of the CNN-LSTM hybrid model as an efficient technique in the classification of business risk and open up promising avenues to explore in the field of finance. With the evolution of the financial environment at such a rapid pace, it is important to focus on efficient ways to classify and forecast business risk with the help of advancements in machine learning techniques such as the one presented in this study.

	precision	recall	f1-score	support
Crash	0.00	0.00	0.00	103
Stable	0.70	1.00	0.82	694
Volatile	0.00	0.00	0.00	198
accuracy			0.70	995
macro avg	0.23	0.33	0.27	995
weighted avg	0.49	0.70	0.57	995





Conclusion and Future Work

The study has offered a holistic CNN-LSTM model solution to the financial health prediction and risk classification of businesses, particularly under small data scenarios. The fact that the CNN-LSTM model incorporates both feature selection and a compact architecture, along with excellent regularization capabilities, helps the proposed solution perform quite effectively and remain interpretable at the same time. This novel solution, apart from helping overcome small data-related challenges, helps ensure that the results of the proposed solution can be interpreted and effectively used.

Moving forward, attention mechanisms could be a direction of future work. These have proved successful in improving the explanation of deep learning models as they enable these models to pay attention to certain parts of the input while predicting. This could prove very valuable in relation to financial forecasts as there might be certain indicators which have more importance at certain times in certain market conditions. Another direction of

future work would be implementing the concept of transfer learning so as to increase performance on less data.

Besides, the incorporation of external macroeconomic indicators into the framework might be expected to offer a wider perspective regarding the factors that determine financial health and business risks. Considering interest rates, inflation, and employment rate variables, the model may arrive at unseen patterns and correlations crucial for appropriate forecasting. In general, these improvements will not only make the model more accurate but also wider-ranging in various fields and thus more effective for risk management and proper decision-making within the context of the financial world.

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